


```

***** MAPDL COMMAND LINE ARGUMENTS *****
BATCH MODE REQUESTED (-b)      = NOLIST
INPUT FILE COPY MODE (-c)      = COPY
DISTRIBUTED MEMORY PARALLEL REQUESTED
  4 PARALLEL PROCESSES REQUESTED WITH SINGLE THREAD PER PROCESS
TOTAL OF 4 CORES REQUESTED
INPUT FILE NAME                 = I:\_ProjectScratch\ScrAD5D\dummy.dat
OUTPUT FILE NAME                = I:\_ProjectScratch\ScrAD5D\solve.out
START-UP FILE MODE              = NOREAD
STOP FILE MODE                  = NOREAD

RELEASE= 2024 R2                BUILD= 24.2      UP20240603  VERSION=WINDOWS x64
CURRENT JOBNAME=file0 10:12:36 OCT 05, 2026 CP=    0.125

PARAMETER _DS_PROGRESS =      999.0000000

/INPUT FILE= ds.dat  LINE=      0

*** NOTE ***
The /CONFIG,NOELDB command is not valid in a distributed memory
parallel solution. Command is ignored.

*GET _WALLSTRT FROM ACTI ITEM=TIME WALL VALUE= 10.2100000

TITLE=
pull out on cement briqq--Static Structural (B5)

ACT Extensions:
  LSDYNA, 2024.2
  5f463412-bd3e-484b-87e7-cbc0a665e474, wbex
/COM, ANSYSMotion, 2024.2
  20180725-3f81-49eb-9f31-41364844c769, wbex

SET PARAMETER DIMENSIONS ON _WB_PROJECTSCRATCH_DIR
TYPE=STRI DIMENSIONS=      248      1      1

PARAMETER _WB_PROJECTSCRATCH_DIR(1) = I:\_ProjectScratch\ScrAD5D\

SET PARAMETER DIMENSIONS ON _WB_SOLVERFILES_DIR
TYPE=STRI DIMENSIONS=      248      1      1

PARAMETER _WB_SOLVERFILES_DIR(1) = I:\pull out on cement briqq_files\dp0\SYS-1\MECH\

SET PARAMETER DIMENSIONS ON _WB_USERFILES_DIR
TYPE=STRI DIMENSIONS=      248      1      1

PARAMETER _WB_USERFILES_DIR(1) = I:\pull out on cement briqq_files\user_files\
--- Data in consistent NMM units. See Solving Units in the help system for more

MPA UNITS SPECIFIED FOR INTERNAL
LENGTH      = MILLIMETERS (mm)
MASS        = TONNE (Mg)
TIME        = SECONDS (sec)
TEMPERATURE = CELSIUS (C)
TOFFSET     = 273.0
FORCE       = NEWTON (N)
HEAT        = MILLIJOULES (mJ)

INPUT UNITS ARE ALSO SET TO MPA

*** MAPDL - ENGINEERING ANALYSIS SYSTEM  RELEASE 2024 R2      24.2      ***
Ansys Mechanical Enterprise Academic Research
01058202 VERSION=WINDOWS x64 10:12:36 OCT 05, 2026 CP=    0.453

pull out on cement briqq--Static Structural (B5)

***** MAPDL ANALYSIS DEFINITION (PREP7) *****
***** Nodes for the whole assembly *****
***** Elements for Body 1 'SYS\Solid' *****
***** Elements for Body 2 'SYS\Solid' *****
***** Elements for Body 3 'Solid' *****
***** Send User Defined Coordinate System(s) *****
***** Set Reference Temperature *****
***** Send Materials *****
***** Create Contact "concrete to concrete" *****
Real Constant Set For Above Contact Is 5 & 4
Create Contact "steel and conc" *****
Real Constant Set For Above Contact Is 7 & 6
Create Contact "steel and comc" *****
Real Constant Set For Above Contact Is 9 & 8
***** Fixed Supports *****
***** Constant Zero Displacement X *****
***** Constant Zero Displacement Z *****

***** ROUTINE COMPLETED ***** CP =      0.766

--- Number of total nodes = 9294
--- Number of contact elements = 3132
--- Number of spring elements = 0
--- Number of bearing elements = 0
--- Number of solid elements = 5476
--- Number of condensed parts = 0
--- Number of total elements = 8608

*GET _WALLBSOL FROM ACTI ITEM=TIME WALL VALUE= 10.2100000
*****
***** SOLUTION *****
*****
***** MAPDL SOLUTION ROUTINE *****

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```

PERFORM A STATIC ANALYSIS
THIS WILL BE A NEW ANALYSIS

LARGE DEFORMATION ANALYSIS

PARAMETER _THICKRATIO =      0.6670000000

USE PRECONDITIONED CONJUGATE GRADIENT SOLVER
CONVERGENCE TOLERANCE = 1.00000E-08
MAXIMUM ITERATION      = NumNode*DofPerNode* 1.0000

CONTACT INFORMATION PRINTOUT LEVEL      1

CHECK INITIAL OPEN/CLOSED STATUS OF SELECTED CONTACT ELEMENTS
AND LIST DETAILED CONTACT PAIR INFORMATION

SPLIT CONTACT SURFACES AT SOLVE PHASE

NUMBER OF SPLITTING TBD BY PROGRAM

DO NOT COMBINE ELEMENT MATRIX FILES (.emat) AFTER DISTRIBUTED PARALLEL SOLUTION

DO NOT COMBINE ELEMENT SAVE DATA FILES (.esav) AFTER DISTRIBUTED PARALLEL SOLUTION

NLDIAG: Nonlinear diagnostics CONT option is set to ON.
Writing frequency : each ITERATION.

DEFINE RESTART CONTROL FOR LOADSTEP LAST
AT FREQUENCY OF LAST AND NUMBER FOR OVERWRITE IS  -1

DELETE RESTART FILES OF ENDSTEP
*****
***** SOLVE FOR LS 1 OF 1 *****
*** Set Displacements ***
CMBLOCK read of NODE component _CM93UY_YP completed

SELECT      COMPONENT
_CM93UY_YP

SPECIFIED CONSTRAINT UY      FOR SELECTED NODES      1 TO      9294 BY      1
REAL= 1.05000000      IMAG= 0.00000000

ALL SELECT      FOR ITEM=NODE COMPONENT=
IN RANGE      1 TO      9294 STEP      1

      9294 NODES (OF      9294 DEFINED) SELECTED BY NSEL COMMAND.
*** Component For All Non-Zero UY Displacements ***

SELECT      COMPONENT
_CM93UY_YP

DEFINITION OF COMPONENT = _DISPNONZEROUY ENTITY=NODE

ALL SELECT      FOR ITEM=NODE COMPONENT=
IN RANGE      1 TO      9294 STEP      1

      9294 NODES (OF      9294 DEFINED) SELECTED BY NSEL COMMAND.

PRINTOUT RESUMED BY /GOP

USE AUTOMATIC TIME STEPPING THIS LOAD STEP

USE      50 SUBSTEPS INITIALLY THIS LOAD STEP FOR ALL DEGREES OF FREEDOM
FOR AUTOMATIC TIME STEPPING:
USE      500 SUBSTEPS AS A MAXIMUM
USE      20 SUBSTEPS AS A MINIMUM

TIME= 1.0000

ERASE THE CURRENT DATABASE OUTPUT CONTROL TABLE.

WRITE ALL ITEMS TO THE DATABASE WITH A FREQUENCY OF NONE
FOR ALL APPLICABLE ENTITIES

WRITE NSOL ITEMS TO THE DATABASE WITH A FREQUENCY OF ALL
FOR ALL APPLICABLE ENTITIES

WRITE RSOL ITEMS TO THE DATABASE WITH A FREQUENCY OF ALL
FOR ALL APPLICABLE ENTITIES

WRITE EANG ITEMS TO THE DATABASE WITH A FREQUENCY OF ALL
FOR ALL APPLICABLE ENTITIES

WRITE ETMP ITEMS TO THE DATABASE WITH A FREQUENCY OF ALL
FOR ALL APPLICABLE ENTITIES

WRITE VENG ITEMS TO THE DATABASE WITH A FREQUENCY OF ALL
FOR ALL APPLICABLE ENTITIES

WRITE STRS ITEMS TO THE DATABASE WITH A FREQUENCY OF ALL
FOR ALL APPLICABLE ENTITIES

WRITE EPEL ITEMS TO THE DATABASE WITH A FREQUENCY OF ALL
FOR ALL APPLICABLE ENTITIES

WRITE EPPL ITEMS TO THE DATABASE WITH A FREQUENCY OF ALL
FOR ALL APPLICABLE ENTITIES

WRITE CONT ITEMS TO THE DATABASE WITH A FREQUENCY OF ALL
FOR ALL APPLICABLE ENTITIES

*GET ANSINTER_ FROM ACTI ITEM=INT VALUE= 0.00000000

*IF ANSINTER_ ( = 0.00000 ) NE
0 ( = 0.00000 ) THEN

*ENDIF

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*** NOTE ***                      CP =          1.078    TIME= 10:12:36
The automatic domain decomposition logic has selected the MESH domain
decomposition method with 4 processes per solution.

***** MAPDL SOLVE      COMMAND *****

*** WARNING ***                  CP =          1.141    TIME= 10:12:36
Element shape checking is currently inactive.  Issue SHPP,ON or
SHPP,WARN to reactivate, if desired.

*** NOTE ***                      CP =          1.203    TIME= 10:12:36
The model data was checked and warning messages were found.
Please review output or errors file (
I:\_ProjectScratch\ScrAD5D\file0.err ) for these warning messages.

*** SELECTION OF ELEMENT TECHNOLOGIES FOR APPLICABLE ELEMENTS ***
--- GIVE SUGGESTIONS AND RESET THE KEY OPTIONS ---

ELEMENT TYPE          1 IS SOLID187. IT IS NOT ASSOCIATED WITH FULLY INCOMPRESSIBLE
HYPERELASTIC MATERIALS. NO SUGGESTION IS AVAILABLE AND NO RESETTING IS NEEDED.

ELEMENT TYPE          2 IS SOLID187. IT IS NOT ASSOCIATED WITH FULLY INCOMPRESSIBLE
HYPERELASTIC MATERIALS. NO SUGGESTION IS AVAILABLE AND NO RESETTING IS NEEDED.

ELEMENT TYPE          3 IS SOLID187. IT IS NOT ASSOCIATED WITH FULLY INCOMPRESSIBLE
HYPERELASTIC MATERIALS. NO SUGGESTION IS AVAILABLE AND NO RESETTING IS NEEDED.

*** MAPDL - ENGINEERING ANALYSIS SYSTEM  RELEASE 2024 R2          24.2    ***
Ansys Mechanical Enterprise Academic Research
01058202 VERSION=WINDOWS x64   10:12:36 OCT 05, 2026 CP=          1.203

pull out on cement briqq--Static Structural (B5)

      S O L U T I O N   O P T I O N S

PROBLEM DIMENSIONALITY. . . . .3-D
DEGREES OF FREEDOM. . . . . UX   UY   UZ
ANALYSIS TYPE . . . . .STATIC (STEADY-STATE)
OFFSET TEMPERATURE FROM ABSOLUTE ZERO . . . . 273.15
NONLINEAR GEOMETRIC EFFECTS . . . . .ON
EQUATION SOLVER OPTION. . . . .PCG
TOLERANCE. . . . .1.00000E-08
NEWTON-RAPHSON OPTION . . . . .PROGRAM CHOSEN
GLOBALLY ASSEMBLED MATRIX . . . . .SYMMETRIC

*** NOTE ***                      CP =          1.266    TIME= 10:12:36
This nonlinear analysis defaults to using the full Newton-Raphson
solution procedure.  This can be modified using the NROPT command.

*** NOTE ***                      CP =          1.266    TIME= 10:12:36
The conditions for direct assembly have been met.  No .emat or .erot
files will be produced.

TRIM CONTACT/TARGET SURFACE

*** WARNING ***                  CP =          2.281    TIME= 10:12:37
Maximum friction coefficient 0.3 is defined in the model.  Switch to
the unsymmetric solver (NROP,UNSYM) instead if convergence difficulty
is encountered.
START TRIMMING SMALL/BONDED CONTACT PAIRS FOR DMP RUN.

      1032 CONTACT ELEMENTS &      534 TARGET ELEMENTS ARE DELETED DUE TO TRIMMING LOGIC.
      3 CONTACT PAIRS ARE REMOVED.

CHECK INITIAL OPEN/CLOSED STATUS OF SELECTED CONTACT ELEMENTS
AND LIST DETAILED CONTACT PAIR INFORMATION

*** WARNING ***                  CP =          2.641    TIME= 10:12:37
Maximum friction coefficient 0.3 is defined in the model.  Switch to
the unsymmetric solver (NROP,UNSYM) instead if convergence difficulty
is encountered.

*** NOTE ***                      CP =          2.641    TIME= 10:12:37
The maximum number of contact elements in any single contact pair is
226, which is smaller than the optimal domain size of 375 elements for
the given number of CPU domains (4).  Therefore, no contact pairs are
being split by the CNCH,DMP logic.

*** NOTE ***                      CP =          2.828    TIME= 10:12:37
Smoothing on certain target nodes (e.g.8017) for pair ID 7 may have
accuracy issue.  Please verify element normal of connected target
elements (e.g.7720 & 7726).

*** NOTE ***                      CP =          3.203    TIME= 10:12:37
Deformable-deformable contact pair identified by real constant set 4
and contact element type 4 has been set up.
Small sliding logic is assumed
Contact algorithm: Augmented Lagrange method
Contact detection at: Gauss integration point
Contact stiffness factor FKN          1.0000
The resulting initial contact stiffness 0.22671E+06
Default penetration tolerance factor FTOLN 0.10000
The resulting penetration tolerance 0.26465
Default opening contact stiffness OPSF will be used.
Default tangent stiffness factor FKT          1.0000
Default elastic slip factor SLTOL 0.50000E-02
The resulting elastic slip tolerance 0.17738E-01
Update contact stiffness at each iteration
Default Max. friction stress TAUMAX 0.10000E+21
Average contact surface length 3.5477
Average contact pair depth 2.6465
Average target surface length 3.5096
Default pinball region factor PINB 0.25000
The resulting pinball region 0.66163
Initial penetration/gap is excluded.
Bonded contact (always) is defined.

```

*** NOTE *** CP = 3.203 TIME= 10:12:37
 Max. Initial penetration 5.719241563E-30 was detected between contact
 element 7502 and target element 7702.

*** NOTE *** CP = 3.203 TIME= 10:12:37
 Deformable-deformable contact pair identified by real constant set 7
 and contact element type 6 has been set up.
 Contact algorithm: Augmented Lagrange method
 Contact detection at: Gauss integration point
 Contact stiffness factor FKN 1.0000
 The resulting initial contact stiffness 0.79329E+06
 Default penetration tolerance factor FTOLN 0.10000
 The resulting penetration tolerance 0.75634E-01
 Max. initial friction coefficient MU 0.30000
 Default tangent stiffness factor FKT 1.0000
 Default elastic slip factor SLTOL 0.10000E-01
 The resulting elastic slip tolerance 0.11737E-01
 Update contact stiffness at each iteration
 Default Max. friction stress TAUMAX 0.10000E+21
 Average contact surface length 1.1737
 Average contact pair depth 0.75634
 Average target surface length 0.99044
 Default pinball region factor PINB 1.0000
 The resulting pinball region 0.75634
 WARNING: Initial penetration is included.

*** NOTE *** CP = 3.203 TIME= 10:12:37
 Max. Initial penetration 7.571507339E-02 was detected between contact
 element 8687 and target element 8139.
 You may move entire target surface by : x= -2.129527784E-02, y=
 6.848860246E-03, z= 7.233516844E-02, to reduce initial penetration.

*** NOTE *** CP = 3.203 TIME= 10:12:37
 Deformable-deformable contact pair identified by real constant set 9
 and contact element type 8 has been set up.
 Contact algorithm: Augmented Lagrange method
 Contact detection at: Gauss integration point
 Contact stiffness factor FKN 1.0000
 The resulting initial contact stiffness 0.78928E+06
 Default penetration tolerance factor FTOLN 0.10000
 The resulting penetration tolerance 0.76018E-01
 Max. initial friction coefficient MU 0.30000
 Default tangent stiffness factor FKT 1.0000
 Default elastic slip factor SLTOL 0.10000E-01
 The resulting elastic slip tolerance 0.10746E-01
 Update contact stiffness at each iteration
 Default Max. friction stress TAUMAX 0.10000E+21
 Average contact surface length 1.0746
 Average contact pair depth 0.76018
 Average target surface length 0.99044
 Default pinball region factor PINB 1.0000
 The resulting pinball region 0.76018
 WARNING: Initial penetration is included.

*** NOTE *** CP = 3.203 TIME= 10:12:37
 Max. Initial penetration 6.707131025E-02 was detected between contact
 element 10071 and target element 9447.
 You may move entire target surface by : x= -4.665647007E-02, y=
 1.897474599E-02, z= 4.429100895E-02, to reduce initial penetration.

*** WARNING *** CP = 3.203 TIME= 10:12:37
 Maximum friction coefficient 0.3 is defined in the model. Switch to
 the unsymmetric solver (NROP,UNSYM) instead if convergence difficulty
 is encountered.

D I S T R I B U T E D D O M A I N D E C O M P O S E R

...Number of elements: 7042
 ...Number of nodes: 9294
 ...Decompose to 4 CPU domains
 ...Element load balance ratio = 1.075

L O A D S T E P O P T I O N S

LOAD STEP NUMBER. 1
 TIME AT END OF THE LOAD STEP. 1.0000
 AUTOMATIC TIME STEPPING ON
 INITIAL NUMBER OF SUBSTEPS 50
 MAXIMUM NUMBER OF SUBSTEPS 500
 MINIMUM NUMBER OF SUBSTEPS 20
 MAXIMUM NUMBER OF EQUILIBRIUM ITERATIONS. 15
 STEP CHANGE BOUNDARY CONDITIONS NO
 STRESS-STIFFENING ON
 TERMINATE ANALYSIS IF NOT CONVERGEDYES (EXIT)
 CONVERGENCE CONTROLS.USE DEFAULTS
 PRINT OUTPUT CONTROLSNO PRINTOUT
 DATABASE OUTPUT CONTROLS
 ITEM FREQUENCY COMPONENT
 ALL NONE
 NSOL ALL
 RSOL ALL
 EANG ALL
 ETMP ALL
 VENG ALL
 STRS ALL
 EPEL ALL
 EPPL ALL
 CONT ALL

SOLUTION MONITORING INFO IS WRITTEN TO FILE=
file.mntr

MAXIMUM NUMBER OF EQUILIBRIUM ITERATIONS HAS BEEN MODIFIED
TO BE, NEQIT = 25, BY SOLUTION CONTROL LOGIC.

*** NOTE *** CP = 4.562 TIME= 10:12:37
The PCG solver has automatically set the level of difficulty for this
model to 2.

***** PRECISE MASS SUMMARY *****

TOTAL RIGID BODY MASS MATRIX ABOUT ORIGIN			Coupled translational/rotational mass		
0.15626E-03	0.0000	0.0000	0.0000	0.19846E-02	0.12046E-09
0.0000	0.15626E-03	0.0000	-0.19846E-02	0.0000	0.21004E-06
0.0000	0.0000	0.15626E-03	-0.12046E-09	-0.21004E-06	0.0000
-----			-----		
			Rotational mass (inertia)		
			0.11211	-0.15981E-06	-0.26636E-05
			-0.15981E-06	0.51818E-01	0.74954E-07
			-0.26636E-05	0.74954E-07	0.96744E-01

TOTAL MASS = 0.15626E-03
The mass principal axes coincide with the global Cartesian axes

CENTER OF MASS (X,Y,Z)= 0.13442E-02 -0.77089E-06 12.700

TOTAL INERTIA ABOUT CENTER OF MASS
0.86910E-01 -0.15981E-06 0.39973E-08
-0.15981E-06 0.26614E-01 0.73424E-07
0.39973E-08 0.73424E-07 0.96744E-01
The inertia principal axes coincide with the global Cartesian axes

*** MASS SUMMARY BY ELEMENT TYPE ***

TYPE	MASS
1	0.779849E-04
2	0.779852E-04
3	0.293845E-06

Range of element maximum matrix coefficients in global coordinates
Maximum = 1295197.34 at element 5406.
Minimum = 905.531054 at element 7504.

TYPE	NUMBER	ENAME	TOTAL CP	AVE CP
1	2519	SOLID187	0.266	0.000105
2	2399	SOLID187	0.281	0.000117
3	558	SOLID187	0.047	0.000084
4	100	CONTA174	0.031	0.000313
5	100	TARGE170	0.000	0.000000
6	208	CONTA174	0.094	0.000451
7	466	TARGE170	0.000	0.000000
8	226	CONTA174	0.141	0.000622
9	466	TARGE170	0.000	0.000000

Time at end of element matrix formulation CP = 4.859375.

ALL CURRENT MAPDL DATA WRITTEN TO FILE NAME= file.rdb

FOR POSSIBLE RESUME FROM THIS POINT

THERE IS TOO MUCH PENETRATION AT 1 CONTACT POINTS OF THE 3D CONTACT ELEMENTS

FORCE CONVERGENCE VALUE = 0.4840E+05 CRITERION= 163.0

Iteration= 5 Ratio= 0.292040 Limit= 1.000000E-08 Wall= 0.0

Iteration= 50 Ratio= 9.283005E-04 Limit= 1.000000E-08 Wall= 0.1

Iteration= 105 Ratio= 2.168598E-06 Limit= 1.000000E-08 Wall= 0.2

DISTRIBUTED PCG SOLVER SOLUTION CONVERGED

DISTRIBUTED PCG SOLVER SOLUTION STATISTICS

NUMBER OF ITERATIONS= 150
NUMBER OF EQUATIONS = 27882
LEVEL OF CONVERGENCE= 1
CALCULATED NORM = 0.95047E-08
SPECIFIED TOLERANCE = 0.10000E-07
TOTAL CPU TIME (sec)= 0.36
TOTAL WALL TIME (sec)= 0.36
TOTAL MEMORY (GB) = 0.02

DISP CONVERGENCE VALUE = 0.5481E-01 CRITERION= 0.2741E-02
EQUIL ITER 1 COMPLETED. NEW TRIANG MATRIX. MAX DOF INC= -0.5481E-01
DISP CONVERGENCE VALUE = 0.4905E-01 CRITERION= 0.2741E-02
LINE SEARCH PARAMETER = 0.8949 SCALED MAX DOF INC = -0.4905E-01
FORCE CONVERGENCE VALUE = 8996. CRITERION= 22.61
EQUIL ITER 2 COMPLETED. NEW TRIANG MATRIX. MAX DOF INC= 0.2883E-01
DISP CONVERGENCE VALUE = 0.2721E-01 CRITERION= 0.2741E-02
LINE SEARCH PARAMETER = 0.9440 SCALED MAX DOF INC = 0.2721E-01
FORCE CONVERGENCE VALUE = 634.2 CRITERION= 17.19
EQUIL ITER 3 COMPLETED. NEW TRIANG MATRIX. MAX DOF INC= 0.4880E-02
DISP CONVERGENCE VALUE = 0.4705E-02 CRITERION= 0.2741E-02
LINE SEARCH PARAMETER = 0.9642 SCALED MAX DOF INC = 0.4705E-02
FORCE CONVERGENCE VALUE = 59.20 CRITERION= 17.23
EQUIL ITER 4 COMPLETED. NEW TRIANG MATRIX. MAX DOF INC= 0.1019E-02
DISP CONVERGENCE VALUE = 0.1019E-02 CRITERION= 0.2741E-02 <<< CONVERGED
LINE SEARCH PARAMETER = 1.000 SCALED MAX DOF INC = 0.1019E-02
FORCE CONVERGENCE VALUE = 7.571 CRITERION= 17.24 <<< CONVERGED
>>> SOLUTION CONVERGED AFTER EQUILIBRIUM ITERATION 4

TYPE	NUMBER	ENAME	TOTAL CP	AVE CP
1	2519	SOLID187	0.188	0.000074
2	2399	SOLID187	0.188	0.000078
3	558	SOLID187	0.047	0.000084
4	100	CONTA174	0.016	0.000156
6	208	CONTA174	0.016	0.000075

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      8      226  CONTA174      0.016  0.000069

*** NODAL LOAD CALCULATION TIMES
TYPE      NUMBER      ENAME      TOTAL CP      AVE CP

      1      2519  SOLID187      0.062  0.000025
      2      2399  SOLID187      0.016  0.000007
      3       558  SOLID187      0.000  0.000000
      4       100  CONTA174      0.000  0.000000
      6       208  CONTA174      0.000  0.000000
      8       226  CONTA174      0.000  0.000000
*** LOAD STEP      1  SUBSTEP      1  COMPLETED.      CUM ITER =      4
*** TIME = 0.200000E-01      TIME INC = 0.200000E-01
*** AUTO STEP TIME: NEXT TIME INC = 0.20000E-01  UNCHANGED

      FORCE CONVERGENCE VALUE = 4516.      CRITERION= 34.48
      DISP CONVERGENCE VALUE = 0.4984E-01 CRITERION= 0.2741E-02
      EQUIL ITER 1 COMPLETED. NEW TRIANG MATRIX. MAX DOF INC= -0.4984E-01
      DISP CONVERGENCE VALUE = 0.3408E-01 CRITERION= 0.2741E-02
      LINE SEARCH PARAMETER = 0.6837 SCALED MAX DOF INC = -0.3408E-01
      FORCE CONVERGENCE VALUE = 1058.      CRITERION= 34.49
      EQUIL ITER 2 COMPLETED. NEW TRIANG MATRIX. MAX DOF INC= -0.1395E-01
      DISP CONVERGENCE VALUE = 0.1270E-01 CRITERION= 0.2741E-02
      LINE SEARCH PARAMETER = 0.9108 SCALED MAX DOF INC = -0.1270E-01
      FORCE CONVERGENCE VALUE = 203.4      CRITERION= 34.50
      EQUIL ITER 3 COMPLETED. NEW TRIANG MATRIX. MAX DOF INC= -0.2182E-02
      DISP CONVERGENCE VALUE = 0.2101E-02 CRITERION= 0.2741E-02 <<< CONVERGED
      LINE SEARCH PARAMETER = 0.9626 SCALED MAX DOF INC = -0.2101E-02
      FORCE CONVERGENCE VALUE = 32.63      CRITERION= 34.50 <<< CONVERGED
      EQUIL ITER 4 COMPLETED. NEW TRIANG MATRIX. MAX DOF INC= 0.2919E-03
      DISP CONVERGENCE VALUE = 0.2919E-03 CRITERION= 0.2741E-02 <<< CONVERGED
      LINE SEARCH PARAMETER = 1.000 SCALED MAX DOF INC = 0.2919E-03
      >>> SOLUTION CONVERGED AFTER EQUILIBRIUM ITERATION 4
*** LOAD STEP      1  SUBSTEP      2  COMPLETED.      CUM ITER =      8
*** TIME = 0.400000E-01      TIME INC = 0.200000E-01
*** AUTO TIME STEP: NEXT TIME INC = 0.30000E-01  INCREASED (FACTOR = 1.5000)

      FORCE CONVERGENCE VALUE = 146.4      CRITERION= 60.42
      DISP CONVERGENCE VALUE = 0.4757E-03 CRITERION= 0.2741E-02 <<< CONVERGED
      EQUIL ITER 1 COMPLETED. NEW TRIANG MATRIX. MAX DOF INC= -0.4757E-03
      DISP CONVERGENCE VALUE = 0.4702E-03 CRITERION= 0.2741E-02 <<< CONVERGED
      LINE SEARCH PARAMETER = 0.9886 SCALED MAX DOF INC = -0.4702E-03
      FORCE CONVERGENCE VALUE = 4.319      CRITERION= 60.41 <<< CONVERGED
      EQUIL ITER 2 COMPLETED. NEW TRIANG MATRIX. MAX DOF INC= 0.1195E-03
      DISP CONVERGENCE VALUE = 0.1195E-03 CRITERION= 0.2741E-02 <<< CONVERGED
      LINE SEARCH PARAMETER = 1.000 SCALED MAX DOF INC = 0.1195E-03
      FORCE CONVERGENCE VALUE = 0.7930      CRITERION= 60.41 <<< CONVERGED
      >>> SOLUTION CONVERGED AFTER EQUILIBRIUM ITERATION 2
*** LOAD STEP      1  SUBSTEP      3  COMPLETED.      CUM ITER =     10
*** TIME = 0.700000E-01      TIME INC = 0.300000E-01
*** AUTO TIME STEP: NEXT TIME INC = 0.45000E-01  INCREASED (FACTOR = 1.5000)

      FORCE CONVERGENCE VALUE = 381.1      CRITERION= 99.39
      DISP CONVERGENCE VALUE = 0.4005E-03 CRITERION= 0.2741E-02 <<< CONVERGED
      EQUIL ITER 1 COMPLETED. NEW TRIANG MATRIX. MAX DOF INC= -0.4005E-03
      DISP CONVERGENCE VALUE = 0.4004E-03 CRITERION= 0.2741E-02 <<< CONVERGED
      LINE SEARCH PARAMETER = 0.9997 SCALED MAX DOF INC = -0.4004E-03
      FORCE CONVERGENCE VALUE = 1.989      CRITERION= 99.34 <<< CONVERGED
      EQUIL ITER 2 COMPLETED. NEW TRIANG MATRIX. MAX DOF INC= -0.1014E-04
      DISP CONVERGENCE VALUE = 0.1014E-04 CRITERION= 0.2741E-02 <<< CONVERGED
      LINE SEARCH PARAMETER = 1.000 SCALED MAX DOF INC = -0.1014E-04
      FORCE CONVERGENCE VALUE = 0.4412      CRITERION= 99.34 <<< CONVERGED
      >>> SOLUTION CONVERGED AFTER EQUILIBRIUM ITERATION 2
*** LOAD STEP      1  SUBSTEP      4  COMPLETED.      CUM ITER =     12
*** TIME = 0.115000      TIME INC = 0.450000E-01
*** AUTO TIME STEP: NEXT TIME INC = 0.50000E-01  INCREASED (FACTOR = 1.1111)

      FORCE CONVERGENCE VALUE = 621.2      CRITERION= 142.9
      DISP CONVERGENCE VALUE = 0.4557E-03 CRITERION= 0.2741E-02 <<< CONVERGED
      EQUIL ITER 1 COMPLETED. NEW TRIANG MATRIX. MAX DOF INC= -0.4557E-03
      DISP CONVERGENCE VALUE = 0.4557E-03 CRITERION= 0.2741E-02 <<< CONVERGED
      LINE SEARCH PARAMETER = 1.000 SCALED MAX DOF INC = -0.4557E-03
      FORCE CONVERGENCE VALUE = 2.219      CRITERION= 142.8 <<< CONVERGED
      >>> SOLUTION CONVERGED AFTER EQUILIBRIUM ITERATION 1
*** LOAD STEP      1  SUBSTEP      5  COMPLETED.      CUM ITER =     13
*** TIME = 0.165000      TIME INC = 0.500000E-01
*** AUTO STEP TIME: NEXT TIME INC = 0.50000E-01  UNCHANGED

      FORCE CONVERGENCE VALUE = 764.0      CRITERION= 186.8
      DISP CONVERGENCE VALUE = 0.4586E-03 CRITERION= 0.2741E-02 <<< CONVERGED
      EQUIL ITER 1 COMPLETED. NEW TRIANG MATRIX. MAX DOF INC= -0.4586E-03
      DISP CONVERGENCE VALUE = 0.4586E-03 CRITERION= 0.2741E-02 <<< CONVERGED
      LINE SEARCH PARAMETER = 1.000 SCALED MAX DOF INC = -0.4586E-03
      FORCE CONVERGENCE VALUE = 3.118      CRITERION= 186.6 <<< CONVERGED
      >>> SOLUTION CONVERGED AFTER EQUILIBRIUM ITERATION 1
*** LOAD STEP      1  SUBSTEP      6  COMPLETED.      CUM ITER =     14
*** TIME = 0.215000      TIME INC = 0.500000E-01
*** AUTO STEP TIME: NEXT TIME INC = 0.50000E-01  UNCHANGED

      FORCE CONVERGENCE VALUE = 901.6      CRITERION= 231.4
      DISP CONVERGENCE VALUE = 0.4621E-03 CRITERION= 0.2741E-02 <<< CONVERGED
      EQUIL ITER 1 COMPLETED. NEW TRIANG MATRIX. MAX DOF INC= -0.4621E-03
      DISP CONVERGENCE VALUE = 0.4621E-03 CRITERION= 0.2741E-02 <<< CONVERGED
      LINE SEARCH PARAMETER = 1.000 SCALED MAX DOF INC = -0.4621E-03
      FORCE CONVERGENCE VALUE = 4.517      CRITERION= 231.0 <<< CONVERGED
      >>> SOLUTION CONVERGED AFTER EQUILIBRIUM ITERATION 1
*** LOAD STEP      1  SUBSTEP      7  COMPLETED.      CUM ITER =     15
*** TIME = 0.265000      TIME INC = 0.500000E-01
*** AUTO STEP TIME: NEXT TIME INC = 0.50000E-01  UNCHANGED

      FORCE CONVERGENCE VALUE = 1068.      CRITERION= 277.2
      DISP CONVERGENCE VALUE = 0.4631E-03 CRITERION= 0.2741E-02 <<< CONVERGED
      EQUIL ITER 1 COMPLETED. NEW TRIANG MATRIX. MAX DOF INC= -0.4631E-03
      DISP CONVERGENCE VALUE = 0.4631E-03 CRITERION= 0.2741E-02 <<< CONVERGED
      LINE SEARCH PARAMETER = 1.000 SCALED MAX DOF INC = -0.4631E-03
      FORCE CONVERGENCE VALUE = 14.51      CRITERION= 276.7 <<< CONVERGED
      >>> SOLUTION CONVERGED AFTER EQUILIBRIUM ITERATION 1
*** LOAD STEP      1  SUBSTEP      8  COMPLETED.      CUM ITER =     16
*** TIME = 0.315000      TIME INC = 0.500000E-01
*** AUTO STEP TIME: NEXT TIME INC = 0.50000E-01  UNCHANGED

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FORCE CONVERGENCE VALUE = 1300. CRITERION= 325.2
DISP CONVERGENCE VALUE = 0.5024E-03 CRITERION= 0.2741E-02 <<< CONVERGED
EQUIL ITER 1 COMPLETED. NEW TRIANG MATRIX. MAX DOF INC= 0.5024E-03
DISP CONVERGENCE VALUE = 0.5024E-03 CRITERION= 0.2741E-02 <<< CONVERGED
LINE SEARCH PARAMETER = 1.000 SCALED MAX DOF INC = 0.5024E-03
FORCE CONVERGENCE VALUE = 43.37 CRITERION= 324.5 <<< CONVERGED
>>> SOLUTION CONVERGED AFTER EQUILIBRIUM ITERATION 1
*** LOAD STEP 1 SUBSTEP 9 COMPLETED. CUM ITER = 17
*** TIME = 0.365000 TIME INC = 0.500000E-01
*** AUTO STEP TIME: NEXT TIME INC = 0.50000E-01 UNCHANGED

FORCE CONVERGENCE VALUE = 1456. CRITERION= 375.6
DISP CONVERGENCE VALUE = 0.4994E-03 CRITERION= 0.2741E-02 <<< CONVERGED
EQUIL ITER 1 COMPLETED. NEW TRIANG MATRIX. MAX DOF INC= 0.4994E-03
DISP CONVERGENCE VALUE = 0.4994E-03 CRITERION= 0.2741E-02 <<< CONVERGED
LINE SEARCH PARAMETER = 1.000 SCALED MAX DOF INC = 0.4994E-03
FORCE CONVERGENCE VALUE = 45.62 CRITERION= 374.6 <<< CONVERGED
>>> SOLUTION CONVERGED AFTER EQUILIBRIUM ITERATION 1
*** LOAD STEP 1 SUBSTEP 10 COMPLETED. CUM ITER = 18
*** TIME = 0.415000 TIME INC = 0.500000E-01
*** AUTO STEP TIME: NEXT TIME INC = 0.50000E-01 UNCHANGED

FORCE CONVERGENCE VALUE = 1768. CRITERION= 430.0
DISP CONVERGENCE VALUE = 0.8628E-03 CRITERION= 0.2741E-02 <<< CONVERGED
EQUIL ITER 1 COMPLETED. NEW TRIANG MATRIX. MAX DOF INC= 0.8628E-03
DISP CONVERGENCE VALUE = 0.8628E-03 CRITERION= 0.2741E-02 <<< CONVERGED
LINE SEARCH PARAMETER = 1.000 SCALED MAX DOF INC = 0.8628E-03
FORCE CONVERGENCE VALUE = 140.7 CRITERION= 428.7 <<< CONVERGED
>>> SOLUTION CONVERGED AFTER EQUILIBRIUM ITERATION 1
*** LOAD STEP 1 SUBSTEP 11 COMPLETED. CUM ITER = 19
*** TIME = 0.465000 TIME INC = 0.500000E-01
*** AUTO STEP TIME: NEXT TIME INC = 0.50000E-01 UNCHANGED

FORCE CONVERGENCE VALUE = 2135. CRITERION= 489.3
DISP CONVERGENCE VALUE = 0.9907E-03 CRITERION= 0.2741E-02 <<< CONVERGED
EQUIL ITER 1 COMPLETED. NEW TRIANG MATRIX. MAX DOF INC= 0.9907E-03
DISP CONVERGENCE VALUE = 0.9907E-03 CRITERION= 0.2741E-02 <<< CONVERGED
LINE SEARCH PARAMETER = 1.000 SCALED MAX DOF INC = 0.9907E-03
FORCE CONVERGENCE VALUE = 212.1 CRITERION= 487.5 <<< CONVERGED
>>> SOLUTION CONVERGED AFTER EQUILIBRIUM ITERATION 1
*** LOAD STEP 1 SUBSTEP 12 COMPLETED. CUM ITER = 20
*** TIME = 0.515000 TIME INC = 0.500000E-01
*** AUTO STEP TIME: NEXT TIME INC = 0.50000E-01 UNCHANGED

FORCE CONVERGENCE VALUE = 2488. CRITERION= 554.6
DISP CONVERGENCE VALUE = 0.1213E-02 CRITERION= 0.2741E-02 <<< CONVERGED
EQUIL ITER 1 COMPLETED. NEW TRIANG MATRIX. MAX DOF INC= 0.1213E-02
DISP CONVERGENCE VALUE = 0.1213E-02 CRITERION= 0.2741E-02 <<< CONVERGED
LINE SEARCH PARAMETER = 1.000 SCALED MAX DOF INC = 0.1213E-02
FORCE CONVERGENCE VALUE = 337.8 CRITERION= 552.3 <<< CONVERGED
>>> SOLUTION CONVERGED AFTER EQUILIBRIUM ITERATION 1
*** LOAD STEP 1 SUBSTEP 13 COMPLETED. CUM ITER = 21
*** TIME = 0.565000 TIME INC = 0.500000E-01
*** AUTO STEP TIME: NEXT TIME INC = 0.50000E-01 UNCHANGED

FORCE CONVERGENCE VALUE = 2734. CRITERION= 626.1
DISP CONVERGENCE VALUE = 0.1028E-02 CRITERION= 0.2741E-02 <<< CONVERGED
EQUIL ITER 1 COMPLETED. NEW TRIANG MATRIX. MAX DOF INC= 0.1028E-02
DISP CONVERGENCE VALUE = 0.1028E-02 CRITERION= 0.2741E-02 <<< CONVERGED
LINE SEARCH PARAMETER = 1.000 SCALED MAX DOF INC = 0.1028E-02
FORCE CONVERGENCE VALUE = 351.0 CRITERION= 623.0 <<< CONVERGED
>>> SOLUTION CONVERGED AFTER EQUILIBRIUM ITERATION 1
*** LOAD STEP 1 SUBSTEP 14 COMPLETED. CUM ITER = 22
*** TIME = 0.615000 TIME INC = 0.500000E-01
*** AUTO STEP TIME: NEXT TIME INC = 0.50000E-01 UNCHANGED

FORCE CONVERGENCE VALUE = 2812. CRITERION= 703.8
DISP CONVERGENCE VALUE = 0.8469E-03 CRITERION= 0.2741E-02 <<< CONVERGED
EQUIL ITER 1 COMPLETED. NEW TRIANG MATRIX. MAX DOF INC= -0.8469E-03
DISP CONVERGENCE VALUE = 0.8469E-03 CRITERION= 0.2741E-02 <<< CONVERGED
LINE SEARCH PARAMETER = 1.000 SCALED MAX DOF INC = -0.8469E-03
FORCE CONVERGENCE VALUE = 345.5 CRITERION= 700.0 <<< CONVERGED
>>> SOLUTION CONVERGED AFTER EQUILIBRIUM ITERATION 1
*** LOAD STEP 1 SUBSTEP 15 COMPLETED. CUM ITER = 23
*** TIME = 0.665000 TIME INC = 0.500000E-01
*** AUTO STEP TIME: NEXT TIME INC = 0.50000E-01 UNCHANGED

FORCE CONVERGENCE VALUE = 3069. CRITERION= 788.8
DISP CONVERGENCE VALUE = 0.8741E-03 CRITERION= 0.2741E-02 <<< CONVERGED
EQUIL ITER 1 COMPLETED. NEW TRIANG MATRIX. MAX DOF INC= -0.8741E-03
DISP CONVERGENCE VALUE = 0.8741E-03 CRITERION= 0.2741E-02 <<< CONVERGED
LINE SEARCH PARAMETER = 1.000 SCALED MAX DOF INC = -0.8741E-03
FORCE CONVERGENCE VALUE = 436.1 CRITERION= 784.2 <<< CONVERGED
>>> SOLUTION CONVERGED AFTER EQUILIBRIUM ITERATION 1
*** LOAD STEP 1 SUBSTEP 16 COMPLETED. CUM ITER = 24
*** TIME = 0.715000 TIME INC = 0.500000E-01
*** AUTO STEP TIME: NEXT TIME INC = 0.50000E-01 UNCHANGED

FORCE CONVERGENCE VALUE = 3607. CRITERION= 882.4
DISP CONVERGENCE VALUE = 0.1050E-02 CRITERION= 0.2741E-02 <<< CONVERGED
EQUIL ITER 1 COMPLETED. NEW TRIANG MATRIX. MAX DOF INC= -0.1050E-02
DISP CONVERGENCE VALUE = 0.1050E-02 CRITERION= 0.2741E-02 <<< CONVERGED
LINE SEARCH PARAMETER = 1.000 SCALED MAX DOF INC = -0.1050E-02
FORCE CONVERGENCE VALUE = 651.4 CRITERION= 876.7 <<< CONVERGED
>>> SOLUTION CONVERGED AFTER EQUILIBRIUM ITERATION 1
*** LOAD STEP 1 SUBSTEP 17 COMPLETED. CUM ITER = 25
*** TIME = 0.765000 TIME INC = 0.500000E-01
*** AUTO STEP TIME: NEXT TIME INC = 0.50000E-01 UNCHANGED

FORCE CONVERGENCE VALUE = 4240. CRITERION= 984.5
DISP CONVERGENCE VALUE = 0.1510E-02 CRITERION= 0.2741E-02 <<< CONVERGED
EQUIL ITER 1 COMPLETED. NEW TRIANG MATRIX. MAX DOF INC= 0.1510E-02
DISP CONVERGENCE VALUE = 0.1510E-02 CRITERION= 0.2741E-02 <<< CONVERGED
LINE SEARCH PARAMETER = 1.000 SCALED MAX DOF INC = 0.1510E-02
FORCE CONVERGENCE VALUE = 752.0 CRITERION= 977.1 <<< CONVERGED
>>> SOLUTION CONVERGED AFTER EQUILIBRIUM ITERATION 1
*** LOAD STEP 1 SUBSTEP 18 COMPLETED. CUM ITER = 26
*** TIME = 0.815000 TIME INC = 0.500000E-01
*** AUTO STEP TIME: NEXT TIME INC = 0.50000E-01 UNCHANGED

FORCE CONVERGENCE VALUE = 4634. CRITERION= 1095.

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***** ROUTINE COMPLETED ***** CP =          42.531

PRINTOUT RESUMED BY /GOP

*GET _WALLDONE FROM ACTI ITEM=TIME WALL VALUE= 10.2211111

PARAMETER _PREPTIME =      0.000000000

PARAMETER _SOLVTIME =      39.000000000

PARAMETER _POSTTIME =      1.000000000

PARAMETER _TOTALTIM =      40.000000000

*GET _DLBRATIO FROM ACTI ITEM=SOLU DLBR VALUE= 1.07460733

*GET _COMBTIME FROM ACTI ITEM=SOLU COMB VALUE= 0.208765100

*GET _SSMODE FROM ACTI ITEM=SOLU SSMM VALUE= 0.000000000

*GET _NDOFS FROM ACTI ITEM=SOLU NDOF VALUE= 25938.0000

*GET _SOL_END_TIME FROM ACTI ITEM=SET TIME VALUE= 1.000000000

*IF _sol_end_time ( = 1.00000 ) EQ
  1.000000 ( = 1.00000 ) THEN

/FCLEAN COMMAND REMOVING ALL LOCAL FILES

*ENDIF
--- Total number of nodes = 9294
--- Total number of elements = 7042
--- Element load balance ratio = 1.07460733
--- Time to combine distributed files = 0.2087651
--- Sparse memory mode = 0
--- Number of DOF = 25938

EXIT MAPDL WITHOUT SAVING DATABASE

NUMBER OF WARNING MESSAGES ENCOUNTERED=      4
NUMBER OF ERROR MESSAGES ENCOUNTERED=      0

+----- M A P D L   S T A T I S T I C S -----+

Release: 2024 R2          Build: 24.2          Update: UP20240603   Platform: WINDOWS x64
Date Run: 10/05/2026    Time: 10:13          Process ID: 13352
Operating System: Windows 11 (Build: 26200)

Processor Model: Intel(R) Xeon(R) Silver 4216 CPU @ 2.10GHz

Compiler: Intel(R) Fortran Compiler Classic Version 2021.9 (Build: 20230302)
          Intel(R) C/C++ Compiler Classic Version 2021.9 (Build: 20230302)
          Intel(R) oneAPI Math Kernel Library Version 2023.1-Product Build 20230303

Number of machines requested      :    1
Total number of cores available   :   32
Number of physical cores available :   16
Number of processes requested     :    4
Number of threads per process requested :    1
Total number of cores requested   :    4 (Distributed Memory Parallel)
MPI Type: INTELMP
MPI Version: Intel(R) MPI Library 2021.11 for Windows* OS

GPU Acceleration: Not Requested

Job Name: file0
Input File: dummy.dat

Core          Machine Name  Working Directory
-----
0             CELAB01-32    I:\_ProjectScratch\ScrAD5D
1             CELAB01-32    I:\_ProjectScratch\ScrAD5D
2             CELAB01-32    I:\_ProjectScratch\ScrAD5D
3             CELAB01-32    I:\_ProjectScratch\ScrAD5D

Latency time from master to core  1 = 6.361 microseconds
Latency time from master to core  2 = 6.081 microseconds
Latency time from master to core  3 = 5.881 microseconds

Communication speed from master to core  1 = 6395.91 MB/sec
Communication speed from master to core  2 = 6716.14 MB/sec
Communication speed from master to core  3 = 6710.34 MB/sec

Total CPU time for main thread      :    40.1 seconds
Total CPU time summed for all threads :    43.1 seconds

Elapsed time spent obtaining a license :    0.5 seconds
Elapsed time spent pre-processing model (/PREP7) :    0.1 seconds
Elapsed time spent solution - preprocessing :    0.7 seconds
Elapsed time spent computing solution :   38.2 seconds
Elapsed time spent solution - postprocessing :    0.2 seconds
Elapsed time spent post-processing model (/POST1) :    0.0 seconds

Equation solver used                :          PCG (symmetric)
Equation solver computational rate   :          9.2 Gflops

Sum of disk space used on all processes :    163.9 MB

Sum of memory used on all processes   :    240.0 MB
Sum of memory allocated on all processes :   2880.0 MB
Physical memory available              :         63 GB
Total amount of I/O written to disk   :         2.3 GB
Total amount of I/O read from disk    :         6.6 GB

+----- E N D   M A P D L   S T A T I S T I C S -----+

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RUN COMPLETED			

Ansys MAPDL 2024 R2	Build 24.2	UP20240603	WINDOWS x64

Database Requested(-db)	1024 MB	Scratch Memory Requested	1024 MB
Max Database Used(Master)	12 MB	Max Scratch Used(Master)	61 MB
Max Database Used(Workers)	1 MB	Max Scratch Used(Workers)	56 MB
Sum Database Used(All)	15 MB	Sum Scratch Used(All)	225 MB

CP Time (sec) =	43.109	Time =	10:13:16
Elapsed Time (sec) =	42.000	Date =	10/05/2026
